FEAT-ZD-48600 · DS-03 Database changes · Completed database changes

Data Science agent · item task.db_changes · source hobasa-data-science-agent/outputs/FEAT-ZD-48600/task.db_changes.md · approved by Developer sim.developer

task.db_changes: FEAT-ZD-48600

Database changes check for the Data Science lane (ap_forecast) of FEAT-ZD-48600 ("Show which overdue invoices belong to vendors that usually pay late"). This is DS-03 Database changes (attempt 1).

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief
Step	DS-03 Database changes (attempt 1)	record:brief
Owner / gate	Developer / automated checks, independent judge, sampled human evaluation	record:brief
Repository	ap_forecast (Data Science lane)	feat.tasks (section 1, TSK-03, TSK-04)
Tasks in scope	TSK-03 (STORY-03, `risk.vendor_risk`), TSK-04 (STORY-04, `explain.build_explanation_request(..., risk=None)`)	feat.tasks (section 1)
Inputs	feat.tasks (b0fd4d2), feat.design (722a98f)	record:brief

1. Result

No database change is required for the Data Science lane, and none is outstanding. The outcome "Required database changes are completed" is met: the set of required changes is empty.

2. Evidence

#	Check	Finding	Source
E-1	Does the design add storage?	The HLD boundary puts "Storage, database, schema migration" out of this design. The data changes are one new record key `paid_date`, one new reject reason `'paid_date'`, an in-memory delay history, risk map and overdue rows, and the CT-1 JSON	feat.design (A1)
E-2	Does the design allow new technology or storage?	DD-1: "No new repository, service, runtime, library or storage." Only the Python 3.9 standard library and Node.js 22 CommonJS	feat.design (A7, DD-1)
E-3	Are the new functions pure?	NF-10: every new function is pure and in-memory, with no I/O, no network and no mutation of inputs	feat.design (A5, NF-10)
E-4	Would storage reopen the platform design?	"any storage or network call from the libraries" is a DES-01 re-entry trigger; the design triggers none of them	feat.design (A8)

#	Check	Finding	Source
E-5	What does TSK-03 change?	New `src/ap_forecast/risk.py`, `__init__.py` (import, `__all__`, version step) and new `tests/test_risk.py`. Rule only: no LLM, no network, no `models.json` read	feat.tasks (TSK-03); feat.design (B2.1, B2.2)
E-6	What does TSK-04 change?	`src/ap_forecast/explain.py` and `tests/test_forecast.py`; `models.json` must not change	feat.tasks (TSK-04); feat.design (B2.1)
E-7	Does ap_forecast use a database today?	No. The repository view states ap_forecast has no database and no migrations exist or are required. A search of the checkout for database, SQL, SQLite, SQLAlchemy, psycopg and migration terms found no match	tool:repo-access; file:ap_forecast/src/ap_forecast/predict.py; file:ap_forecast/src/ap_forecast/explain.py
E-8	What persistent data does ap_forecast read?	Only `models.json`, a JSON configuration file read by `explain.load_config()` (provider, model, `max_tokens`). It is configuration, not a database, and it is unchanged by this feature	file:ap_forecast/src/ap_forecast/explain.py; feat.design (B2.1)
E-9	Is `predict_delay_days` stateful?	No. It computes the rounded mean of the given history in memory, with no I/O	file:ap_forecast/src/ap_forecast/predict.py
E-10	Is the `.codegraph/` database part of the product?	No. It is a local code-intelligence index; its `.gitignore` ignores everything except itself, so it is never committed and is not an application database	file:ap_forecast/.codegraph/.gitignore

3. Database change list

Change	Type	Status
None	—	Not required (sections 1 and 2)

No schema, table, index, migration script, seed data, grant or DBA ticket is needed for TSK-03 or TSK-04. Nothing waits on infra or DBAs for this lane.

4. Impact on the next steps

- DS-04 (components) and DS-05 (coding) can proceed without a database prerequisite.
- If a later change to the design introduces storage for delay history or risk levels, that is a DES-01 re-entry (feat.design A8), and DS-03 must be re-run.

5. Out of scope for this lane

The other lanes (ap_pipeline, ap_aging, ap_portal) are not checked here. The design states the same position for the whole feature (no storage, database or schema migration; feat.design A1, DD-1), but each lane confirms its own repository in its own step.

6. Open questions

None. No blocking question.

Tool results (recorded by the spine from the real tools)

[Tool result] [ap_forecast] No database in this repository's architecture; no migrations required